Evaluating the Contribution of Road Network Topology to Surface Urban Heat Island Estimation in Davao City Using Graph Neural Networks

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CHAPTER I
INTRODUCTION

Urbanization and changes in land use have intensified environmental challenges in cities facing the combined pressures of development and climate change. As urban populations grow, more residents are exposed to elevated temperatures, increasing the need for locally appropriate heat-mitigation measures. A global assessment of future urban temperatures emphasizes the potential role of nature-based solutions in limiting these impacts [1]. Satellite evidence also shows that urban surface warming can exceed warming in surrounding areas: an analysis of more than 2,000 city clusters found that the mean surface warming trend in urban cores was 29% greater than in rural backgrounds during 2002-2021 [2]. The combined influence of urban development and climate change has been investigated through complementary methods. A study of Lagos separated urbanization-related and broader climatic contributions to changes in urban surface temperature, demonstrating the importance of distinguishing local land-cover effects from background warming [3]. Scenario-based research in Tehran also examined urbanization and climate change in relation to urban heat and thermal comfort [4].

At the global scale, projections integrating future urban expansion and climate change indicate that urbanization can add to local surface warming through 2100 [5]. Land surface temperature (LST) describes the thermal condition of the ground, vegetation, roofs, and other surfaces as estimated from thermal infrared observations [6]. It differs from the temperature of the overlying air. A surface urban heat island (SUHI) occurs when urban surfaces are warmer than a specified surrounding non-urban background; SUHI intensity expresses this difference rather than the absolute surface temperature [7]. Consequently, high LST alone does not establish the magnitude of SUHI intensity. The reference surface must be defined explicitly because differences in rural land cover can change the resulting urban-rural temperature contrast [8]. Research comparing crowdsourced air temperatures with satellite observations further shows that surface and air-temperature heat islands can differ in magnitude and their relationships with environmental characteristics [9]. The relevance of urban thermal information extends to adaptation planning. A review of resilience to weather and climate extremes identifies infrastructure, socioeconomic conditions, and governance as interacting influences on urban adaptive capacity [10]. Research on urban heat-risk management similarly emphasizes differentiated vulnerability, participation, uncertainty, and

the sequencing of adaptation actions [11]. Consequently, identifying warmer urban surfaces is a useful component of environmental assessment, but deciding where and how to intervene requires evidence about local conditions and the people affected. This distinction helps define the contribution of a surface-temperature model within a broader planning process.

The intensification of urban heat is particularly critical in Southeast Asia, where high humidity and accelerating development expose growing populations to severe thermal risks [12], [13]. Regional projections indicate that the combined pressures of climate change and urban expansion will significantly elevate local surface temperatures [14], [15]. While extensive research has characterized broad surface thermal variations across tropical cities [16],[17], there remains a critical need for high-resolution spatial modeling to inform localized urban planning. Consequently, accurate, street-level intra-urban heat mapping serves as an essential first step before communities can evaluate and implement practical cooling interventions [18], [19].

In the Philippines, long-term meteorological records provide evidence of changes in temperature extremes. A national assessment analyzed daily precipitation and near-surface air temperature for 1951-2010 [20]. A subsequent analysis of observations from 34 meteorological stations for 1951-2018 identified differences in temperature trends between urban and rural stations, with stronger warming in indices associated with minimum temperature [21]. These atmospheric observations establish a national context, but they should not be treated as direct measurements of satellite-derived SUHI intensity or as proof that every observed temperature change is caused by urbanization. Philippine remote-sensing studies provide more direct evidence of surface thermal variation. An analysis of Metropolitan Manila found that increasing built-up indicators and declining vegetation indicators were associated with higher LST in Metropolitan Manila between 2001 and 2019 [22]. A study of Manila City combined satellite and meteorological information to assess intra-urban thermal variation in Manila City and locate areas for further intervention assessment [23]. Beyond the capital region, research examined land-cover changes in the Cagayan de Oro River basin from 2003 to 2020 and reported higher mean LST over built-up areas and bare land than over forest, mangroves, and water bodies [24]. In fact, a localized heat-risk assessment in Quezon City integrates environmental exposure and socioeconomic vulnerability in a weighted index, illustrating how densely developed and vegetation-poor areas can overlap with socially vulnerable communities [25].

Atmospheric modeling offers complementary evidence about the Philippine urban environment. A numerical simulation study used the Weather Research and Forecasting model coupled with an urban canopy model to examine urban land cover and anthropogenic heat during extreme heat events in Metro Manila [26]. A related study evaluated urban canopy model configurations for air temperature and relative humidity during a high-heat event, emphasizing the importance of appropriate urban morphology information [27]. These studies concern atmospheric processes and do not establish that heat travels along road-network connections. At the national scale, a heat-health assessment combined MODIS-derived thermal information with population data for a heat-health assessment [28].

These thermal pressures are also evident in Davao City. A local study mapped the occurrence of hot and cold surface-temperature locations from 1994 to 2019 and examined associations with vegetation, built-up indicators, albedo, elevation, and population density. Its quantile-regression results identified the Normalized Difference Built-up Index (NDBI) as the most significant predictor at the 50th, 75th, and 90th LST quantiles, while other predictors were more important at lower quantiles [29]. This variation cautions against assuming that a single environmental factor explains thermal conditions throughout the city. Another local investigation reported higher surface temperatures in densely built-up, vegetation-deficient parts of Davao than in greener areas [30]. These local studies establish a basis for further surface-temperature modeling, but they do not determine whether road-network connectivity adds predictive information beyond the environmental attributes observed at each location. Additional Landsat-based research in Davao City for 2018-2022 reports a positive association between NDBI and LST and a negative association between NDVI and LST [31].

Satellite remote sensing provides spatially extensive information for examining these local differences, subject to cloud cover, retrieval uncertainty, and the spatial and temporal characteristics of the observations [6], [7]. Machine-learning approaches used in urban heat research include Random Forest (RF), convolutional neural networks (CNNs), and other neural architectures [32]. A further comprehensive review examines the range of machine-learning methods used for urban heat-island assessment [33]. However, predictive performance depends on how the data, spatial relationships, and evaluation procedure are represented. A review of environmental spatial machine learning identifies several ways to account for spatial autocorrelation, including spatial covariates and training-test partitioning, and finds no single approach that is appropriate for every application [34].

Empirical comparisons provide a reason to retain a conventional benchmark. Fine-scale urban heat modeling has compared Random Forest with multiple linear regression using air temperature and a broad set of environmental and urban predictors [35]. Although the response variable differs from satellite LST, the study demonstrates the relevance of evaluating nonlinear models against alternatives under a defined task. An open-source UHI-mapping framework for Lagos uses a U-Net architecture with satellite and urban geospatial inputs [36], illustrating a raster-based approach to a related thermal problem.

Although Random Forest can capture nonlinear relationships between environmental predictors and LST, a conventional tabular implementation does not explicitly aggregate information through road-network connections unless such information is encoded in its inputs. Standard raster CNNs represent local spatial context through grid neighborhoods; those neighborhoods do not explicitly distinguish locations connected by a road from locations that are merely close in geographic space. A road graph provides a different representation: intersections and intermediate road points become nodes, and connecting road segments define their adjacency. Street-graph research demonstrates the usefulness of network-based units for characterizing urban morphology, although that work uses geometric information beyond connectivity alone [37]. In this study, road adjacency is treated as a candidate relational representation whose predictive contribution must be tested. It is not assumed to be a physical heat-transfer pathway or a complete description of street-canyon geometry.

A graph representation also requires a clear interpretation of its edges. Research on regional surface heat islands has combined morphological spatial pattern analysis and graph-based connectivity measures to evaluate the configuration of thermal patches [38]. Such a thermal-landscape graph differs from a street graph because its connections are defined for a different analytical purpose. This distinction matters when adapting graph methods: an adjacency matrix is a modeling choice, and evidence obtained with thermal patches cannot by itself validate road adjacency. In the proposed model, connected road locations provide neighborhoods for information aggregation, with their usefulness assessed through prediction errors.

Graph neural networks (GNNs) offer mechanisms for learning from both node attributes and the relationships specified by a graph [39]. Environmental applications provide relevant methodological precedents. One study developed graph-based spatiotemporal interpolation using a Voronoi-adjacency representation [40]. Another study applied a GCN

framework to daytime and nighttime SUHI modeling in Hong Kong, integrating built and natural urban features [41][45]. A recent review examined graph construction and spatial-temporal learning across several urban environmental applications [42][46]. These studies support investigating graph-based thermal estimation, but results obtained with other graph definitions, environmental variables, and locations cannot establish the benefit of Davao City's road topology in advance.

While recent environmental applications have successfully utilized Graph Neural Networks (GNNs) for tasks such as spatiotemporal interpolation and road-surface temperature forecasting [40], [42], [43], these models are primarily designed for short-term atmospheric forecasting. Estimating annual satellite-derived LST requires a different temporal approach. Consequently, while these graph architectures provide a methodological precedent, their results motivate further investigation without justifying claims about subpixel road temperatures or physical heat movement in the present study.

The present study will represent road-associated locations in Davao City using an OpenStreetMap (OSM) network acquired and processed with OSMnx, which provides methods for constructing and analyzing street graphs [44]. Nodes will include intersections and intermediate points along road segments. Each node will be associated with historical LST, the Normalized Difference Vegetation Index (NDVI), and NDBI. NDVI and NDBI will be treated as spectral indicators of vegetation and built-up characteristics, respectively; NDBI will not be interpreted as a direct measurement of impervious-surface density. Road connectivity will allow the model to combine information from connected locations. The hypothesis is that this relational information may contribute to estimation after accounting for the historical environmental features at each node. Combining OSM with satellite and environmental information has also been explored in a multi-domain urban-planning framework that addresses land use, building footprints, traffic, and air quality [45]. This provides a precedent for integrating volunteered geographic information with remotely sensed inputs, while the present study has a narrower thermal-estimation purpose. OSM coverage and the alignment of road geometry with satellite observations will consequently be documented as properties of the input data. A more detailed road graph will not be treated as evidence of equally detailed thermal measurements.

The proposed architecture combines a Long Short-Term Memory (LSTM) encoder with GraphSAGE neighborhood-aggregation layers. LSTM provides the temporal sequence-learning component [46], while GraphSAGE provides a method for learning node representations by aggregating neighborhood information [47]. In this chapter, the specific model is called LSTM-GraphSAGE to distinguish its architecture from the broader GNN family. Annual median environmental features from a continuous six-year historical sequence (2019–2024) will be used to estimate the 2025 LST associated with each node. SUHI intensity will then be derived for urban nodes by subtracting a common, independently specified non-urban reference LST for 2025. This 7-year historical target-year experiment evaluates spatial estimation using a robust multi-year sequence; it does not establish operational short-term forecasting or reliable prediction under future infrastructure changes.

The methodological gap is whether explicitly representing road-network connectivity improves estimation beyond what can be learned from the same historical satellite-derived features without information exchange between locations. Comparing LSTM–GraphSAGE with Random Forest alone would not isolate this contribution because both the architecture and the use of adjacency would change. The study will therefore compare the road-connected model with a matched configuration in which all edges between distinct nodes are removed, and only self-loops are retained. In this configuration, the LSTM encoder and GraphSAGE layers remain, but each node processes only its own historical features and representation, without receiving information from other nodes. The resulting model operates as a shared node-wise predictor at the same road-associated locations. Both neural configurations will use the same input features, layer dimensions, and training procedure. A Random Forest model using the same historical predictors will provide a separate conventional benchmark. All configurations will be assessed on the same held-out observations.

Through these comparisons, the study will evaluate both overall estimation performance and the added predictive value of the chosen road-network representation. Its outputs will include estimated LST, derived SUHI intensity at urban road-network nodes, and maps of locations with elevated estimated SUHI intensity. These outputs are intended to support the interpretation of surface thermal variation within the defined urban study area of Davao City. Neither improved performance nor the ability to identify hotspots will be assumed before evaluation.

Research Questions

This study seeks to evaluate whether road-network topology contributes useful information to the estimation and mapping of satellite-derived SUHI intensity in Davao City. Specifically, it will address the following questions:

1. How accurately can the proposed LSTM–GraphSAGE model, integrating OSM road-network topology and historical satellite-derived environmental features, estimate 2025 land surface temperature (LST) at urban road-network nodes in Davao City?
2. What change in estimation performance results from incorporating road-network connectivity compared with (a) a matched configuration where inter-node connections are disabled, and (b) a spatial-proximity graph (e.g., K-Nearest Neighbors) that connects nodes based purely on geographic distance?
3. How does the predictive performance of the proposed model compare with a Random Forest regression baseline using the same historical predictors and held-out observations, based on root mean square error (RMSE), mean absolute error (MAE), and the coefficient of determination (R^2)?

Significance of the Study

This study will provide spatially explicit information on SUHI intensity and an empirical assessment of road-network topology as a representation for thermal estimation. Its potential contributions concern urban planning, local environmental assessment, and reproducible geospatial machine learning.

GeoAI and Computer Science. The comparison between road-connected and connectivity-disabled configurations will examine the predictive value of an explicit network representation while controlling the principal architectural differences. The separate Random Forest benchmark will assess the complete model against a conventional machine-learning approach. A finding that road connectivity provides little or no improvement would also be informative because it would clarify the limits of this representation for the selected data and task.

Remote Sensing and Urban Climate Research. The study will relate satellite-derived thermal and spectral information to road-associated locations while maintaining the distinction between LST and reference-based SUHI intensity. Documenting

the reference-area definition, imagery coverage, and spatial support of the estimates will help make the results interpretable and comparable with subsequent local studies.

Urban Planning and Engineering. The resulting maps may help the City Planning and Development Office and the City Engineer's Office identify road-associated locations with elevated estimated SUHI intensity for further site assessment. Such information may inform the consideration of shading, vegetation, and surface-material interventions alongside field observations and engineering requirements. The study will not determine the optimal intervention or demonstrate the cooling effectiveness of cool pavements, permeable materials, or urban greening.

Local Government and Environmental Management. The study may provide supporting spatial evidence for discussions of land-use planning, zoning, and urban greening priorities. The maps can indicate where further environmental investigation is warranted, but they will not constitute a heat-health warning system. Decisions involving disaster response or public health would require additional meteorological, exposure, and vulnerability information.

Open and Reproducible Geospatial Modeling. The use of OSM, Landsat data, and documented processing procedures will support reproducibility. Recording acquisition dates, quality-screening rules, graph-construction decisions, model settings, data partitions, and evaluation procedures will enable other researchers to examine the workflow and assess its suitability for comparable urban contexts.

Scope and Delimitations

The geographic extent of road-network acquisition will follow the Davao City administrative boundary obtained from OSM. This acquisition extent includes urban, agricultural, rural, and mountainous areas; however, citywide acquisition does not mean that every road node represents an urban SUHI observation. The principal estimation and interpretation will focus on nodes within the continuously built-up urban area, which will be operationally defined using an objective, reproducible land-cover classification dataset (e.g., ESA WorldCover) to eliminate manual boundary bias. Urban evaluation locations and non-urban reference surfaces will be distinguished through documented geographic and land-cover masks. Areas will not be designated urban, excluded, or retained solely to obtain a more connected graph. The exact masks, road-type filters, and treatment of disconnected components will be specified in the methodology and reported with the resulting coverage.

The study will use Landsat 8 Collection 2 Level-2 surface reflectance and surface temperature products for 2019-2025 to extract surface infrastructure indices, vegetation indices, and annual median thermal observations. The USGS surface temperature product provides the thermal observations, while NDVI and NDBI will be computed from the corresponding reflectance data using the appropriate product scaling and quality information [56]. Cloud, cloud-shadow, and invalid observations will be screened before constructing annual median composites. Historical LST, NDVI, and NDBI from 2019 to 2024 will form the sequential model inputs, and the 2025 LST composite will provide the target observations... To preserve a mathematically consistent 16-day orbital sampling frequency across all seven years and prevent artificial warming biases, multisensor fusion with Landsat 9 is explicitly excluded from the analysis.

The model will estimate LST, from which urban-node SUHI intensity will be derived. For an urban node i and target year t, the operational relationship is SUHI(i,t) = LST(i,t) - LSTref(t), where LSTref(t) is the median of valid annual-composite LST pixels within the designated non-urban reference mask for the same year. The estimated SUHI will use the model-estimated urban LST and the same reference value used for the satellite-derived comparison. The reference mask will be defined independently of model prediction errors and will retain non-urban reference pixels even when they are outside the modeled urban graph. Its land-cover composition and elevation comparability with the urban area will be documented, and water surfaces will not be pooled into the terrestrial rural reference. Because reference selection affects SUHI interpretation, the reference boundary and eligibility criteria must be fixed before final evaluation [8]. This procedure produces a reference-based surface-temperature contrast, not an independently measured air-temperature heat island.

The road graph will contain intersections and intermediate points along retained road segments. Node densification changes the locations at which features are assigned and predictions are reported; it does not create finer independent thermal observations. Spatial detail remains constrained by the native and processed support of the Landsat observations. Accordingly, references to the street-network level describe the placement of estimates on a road graph, not direct measurement of pavement temperature or pedestrian-level microclimate. The road-network representation will also depend on OSM completeness, positional accuracy, attribute quality, and the date of extraction. An OSM network snapshot will not be assumed to reproduce all road changes occurring during 2019-2025.

The comparative analysis will evaluate four configurations: (1) LSTM–GraphSAGE with OSM road adjacency, (2) a matched configuration without information exchange between distinct nodes, (3) a matched configuration utilizing a spatial-proximity graph (e.g., K-Nearest Neighbors) based on geographic distance, and (4) Random Forest regression. To ensure the matched configurations isolate the contributions of road and spatial adjacency without altering the underlying neural architecture, 'without information exchange' is defined computationally by restricting the adjacency matrix exclusively to self-loops (i.e. aggregating only self-information). These comparisons will assess the contribution of road-network connectivity and the performance of the proposed approach relative to a conventional machine-learning benchmark. Evaluation will account for spatial dependence and shared satellite observation support [38]. The model configurations, training procedures, and spatial train/validation/test partitioning strategy will be specified in Chapter 3.

Predictive performance will be evaluated for LST using RMSE, MAE, and R^2. SUHI intensity will be derived from estimated LST using a designated target-year non-urban reference and used primarily for spatial interpretation and mapping. It will not be treated as an independently predicted target or subjected to a separate, redundant assessment of prediction error. Maps of estimated LST and derived SUHI intensity will present their spatial distribution at urban road-network nodes rather than constitute a separately evaluated hotspot-identification system.

The study will use satellite observations covering 2019–2025, with conclusions restricted to the input periods and target years evaluated. Annual aggregation does not support conclusions about seasonal behavior, diurnal thermal patterns, or short-term heat-event forecasting. The analysis excludes ground-based air-temperature prediction, thermal comfort, heat index, human heat exposure, and health-risk estimation. It will not simulate physical heat transport along roads, quantify the causal effects of individual road connections, or predict the cooling outcomes of infrastructure changes. Its conclusions will be limited to comparative LST estimation performance and mapped surface thermal conditions under the documented observation and SUHI reference definitions.

References

[1] M. Esperon-Rodriguez *et al.*, “Urban heat in global cities and the role of nature-based solutions in mitigating future climate risks,” *Environ. Res. Clim.*, vol. 4, no. 2, p. 023001, Apr. 2025, doi: 10.1088/2752-5295/adcb61.

[2] Z. Liu *et al.*, “Surface warming in global cities is substantially more rapid than in rural background areas,” *Commun. Earth Environ.*, vol. 3, no. 1, p. 219, Sep. 2022, doi: 10.1038/s43247-022-00539-x.

[3] L. Guo, L. Di, C. Zhang, L. Lin, F. Chen, and A. Molla, “Evaluating contributions of urbanization and global climate change to urban land surface temperature change: a case study in Lagos, Nigeria,” *Sci. Rep.*, vol. 12, no. 1, p. 14168, Aug. 2022, doi: 10.1038/s41598-022-18193-w.

[4] M. Zarghamipour and H. Malakooti, “The projected effects of urbanization and climate change on urban Heat Island and thermal comfort over the Tehran metropolitan,” *Sci. Total Environ.*, vol. 992, p. 179955, Aug. 2025, doi: 10.1016/j.scitotenv.2025.179955.

[5] S. Liu, X. Li, Z. Shi, M. Geng, G. Yu, and T. Hu, “Urbanization is projected to increase local surface temperature by 2100,” *Commun. Earth Environ.*, vol. 6, no. 1, p. 988, Dec. 2025, doi: 10.1038/s43247-025-02947-1.

[6] Z.-L. Li *et al.*, “Satellite Remote Sensing of Global Land Surface Temperature: Definition, Methods, Products, and Applications,” *Rev. Geophys.*, vol. 61, no. 1, p. e2022RG000777, 2023, doi: 10.1029/2022RG000777.

[7] D. Zhou *et al.*, “Satellite Remote Sensing of Surface Urban Heat Islands: Progress, Challenges, and Perspectives,” *Remote Sens.*, vol. 11, no. 1, p. 48, Jan. 2019, doi: 10.3390/rs11010048.

[8] X. Yang and L. Yao, “Reexamining the relationship between surface urban heat island intensity and annual precipitation: Effects of reference rural land cover,” *Urban Clim.*, vol. 41, p. 101074, Jan. 2022, doi: 10.1016/j.uclim.2021.101074.

[9] Z. S. Venter, T. Chakraborty, and X. Lee, “Crowdsourced air temperatures contrast satellite measures of the urban heat island and its mechanisms,” *Sci. Adv.*, vol. 7, no. 22, p. eabb9569, May 2021, doi: 10.1126/sciadv.abb9569.

[10] A. A. Akinsanola *et al.*, “A review of urban resilience to weather and climate extremes,” *City Built Environ.*, vol. 3, no. 1, p. 24, Nov. 2025, doi: 10.1007/s44213-025-00063-6.

[11] C. Singh *et al.*, “Applying recent advances in climate adaptation research to urban heat risk management,” *WIREs Clim. Change*, vol. 15, no. 5, p. e901, Sep. 2024, doi: 10.1002/wcc.901.

[12] M. C. Wong, J. Wang, X. Zhi, and L. Dong, “A 1940-2020 spatiotemporal analysis of thermal discomfort days in Southeast Asian countries,” *Environ. Res. Commun.*, vol. 6, no. 10, p. 101009, Oct. 2024, doi: 10.1088/2515-7620/ad810b.

[13] X. Sun, F. Ge, Y. Fan, S. Zhu, and Q. Chen, “Will population exposure to heat extremes intensify over Southeast Asia in a warmer world?,” *Environ. Res. Lett.*, vol. 17, no. 4, p. 044006, Mar. 2022, doi: 10.1088/1748-9326/ac48b6.

[14] Y. L. Ng, M. H. Lim, Y. F. Huang, and Y. L. Lee, “Thermal intensification and urban land cover change in Southeast Asian capitals: over three decades of satellite observations and trend analysis,” *Npj Urban Sustain.*, vol. 6, no. 1, p. 11, Dec. 2025, doi: 10.1038/s42949-025-00315-8.

[15] S. Manimaran *et al.*, “Widespread heat stress will become the norm in a warming Southeast Asia,” *Sci. Rep.*, vol. 15, no. 1, p. 45799, Dec. 2025, doi: 10.1038/s41598-025-28817-6.

[16] C. T. Nguyen *et al.*, “How do disparate urbanization and climate change imprint on urban thermal variations? A comparison between two dynamic cities in Southeast Asia,” *Sustain. Cities Soc.*, vol. 82, p. 103882, Jul. 2022, doi: 10.1016/j.scs.2022.103882.

[17] Y. Zhu *et al.*, “Thermal changes along the urban-rural continuums in Southeast Asia,” *Environ. Res. Lett.*, vol. 20, no. 5, p. 054051, May 2025, doi: 10.1088/1748-9326/adcad2.

[18] K. Z. Tun *et al.*, “Cooling the cities: A comprehensive review of urban heat island mitigation strategies in Southeast Asia,” *Hum. Settl. Sustain.*, vol. 1, no. 2, pp. 91–102, Jun. 2025, doi: 10.1016/j.hssust.2025.05.002.

[19] L. L. Delina, H. Pagkalinawan, and S. F. A. Macagba, “Integrating Satellite-Based and Participatory Spatial Analyses for Local, Contextualised, and Nuanced Extreme Heat Hotspot Mapping in Overheated Urban Southeast

Asia," *Sustain. Dev.*, vol. 34, no. S2, pp. 1015–1033, 2026, doi: 10.1002/sd.70329.

[20] T. A. Cinco, R. G. de Guzman, F. D. Hilario, and D. M. Wilson, "Long-term trends and extremes in observed daily precipitation and near surface air temperature in the Philippines for the period 1951–2010," *Atmos. Res.*, vol. 145–146, pp. 12–26, Aug. 2014, doi: 10.1016/j.atmosres.2014.03.025.

[21] J. A. Manalo *et al.*, "The effect of urbanization on temperature indices in the Philippines," *Int. J. Climatol.*, vol. 42, no. 2, pp. 850–867, 2022, doi: 10.1002/joc.7276.

[22] K. J. Almadrones-Reyes and N. H. A. Dagamac, "Land-use/land cover change and land surface temperature in Metropolitan Manila, Philippines using landsat imagery," *GeoJournal*, vol. 88, no. 2, pp. 1415–1426, Apr. 2023, doi: 10.1007/s10708-022-10701-9.

[23] M. A. Purio, T. Yoshitake, and M. Cho, "Assessment of Intra-Urban Heat Island in a Densely Populated City Using Remote Sensing: A Case Study for Manila City," *Remote Sens.*, vol. 14, no. 21, p. 5573, Jan. 2022, doi: 10.3390/rs14215573.

[24] R. A. Galvez, H. A. Roa-Quiaoit, F. L. Dagoc, J. Guihawan, and P. Suson, "Modeling the influence of land cover dynamics on spatio-temporal variations in land surface temperature in Cagayan de Oro River basin, Mindanao, Philippines," *Model. Earth Syst. Environ.*, vol. 10, no. 1, pp. 899–912, Feb. 2024, doi: 10.1007/s40808-023-01834-y.

[25] A. C. Treyes, "When the city heats up: mapping urban heat risks through environmental and socioeconomic factors in Quezon City, Philippines," *Environ. Res. Clim.*, vol. 4, no. 2, p. 025012, Jun. 2025, doi: 10.1088/2752-5295/addd41.

[26] A. M. T. Magnaye and H. Kusaka, "Potential effect of urbanization on extreme heat events in Metro Manila Philippines using WRF-UCM," *Sustain. Cities Soc.*, vol. 110, p. 105584, Sep. 2024, doi: 10.1016/j.scs.2024.105584.

[27] R. G. J. P. Bilang, A. C. Blanco, J. A. S. Santos, and L. M. P. Olaguera, "Simulation of Urban Heat Island during a High-Heat Event Using WRF Urban Canopy Models: A Case Study for Metro Manila," *Atmosphere*, vol. 13, no. 10, p. 1658, Oct. 2022, doi: 10.3390/atmos13101658.

[28] D. Vergara and A. Blanco, “SURFACE URBAN HEAT ISLANDS AND RELATED HEALTH RISK IN THE PHILIPPINES: A GEOSPATIAL ASSESSMENT USING MODIS DATA,” *Int. Arch. Photogramm. Remote Sens. Spat. Inf. Sci.*, vol. XLVIII-4/W6-2022, pp. 451–456, Feb. 2023, doi: 10.5194/isprs-archives-XLVIII-4-W6-2022-451-2023.

[29] M. M. Tinoy, A. U. Novero, K. P. Landicho, A. B. Baloloy, and A. C. Blanco, “URBAN EFFECTS ON LAND SURFACE TEMPERATURE IN DAVAO CITY, PHILIPPINES,” *Int. Arch. Photogramm. Remote Sens. Spat. Inf. Sci.*, vol. XLII-4-W19, pp. 433–440, Dec. 2019, doi: 10.5194/isprs-archives-XLII-4-W19-433-2019.

[30] M. Batican and J. B. Paragamac, “Spatiotemporal Analysis Of Urban Land Surface Temperature (Ulst) In Davao City,” *Res. Front. Int. J. Soc. Sci. Technol.*, vol. 1, no. 2, pp. 129–150, Nov. 2025, doi: 10.55990/20250009.

[31] R. Dumdum and J. Paragamac, “GIS-Based Spatio-Temporal Analysis of Vegetation Dynamics, Urban Expansion, and Land Surface Temperature in Davao City,” *Int. J. Innov. Sci. Res. Technol.*, p. 2112, Jul. 2026, doi: 10.38124/ijisrt/26jul851.

[32] R. Snaiki and A. Merabtine, “Recent advances on machine learning techniques for urban heat island applications: a review and new horizons,” *Sustain. Cities Soc.*, vol. 134, p. 106943, Nov. 2025, doi: 10.1016/j.scs.2025.106943.

[33] A. Gaur and C. Deb, “Machine learning methods and approaches for Urban Heat Island (UHI) assessment: A comprehensive review,” *Renew. Sustain. Energy Rev.*, vol. 234, p. 116903, Jul. 2026, doi: 10.1016/j.rser.2026.116903.

[34] M. Jemeļjanova, A. Kmoch, and E. Uuemaa, “Adapting machine learning for environmental spatial data - A review,” *Ecol. Inform.*, vol. 81, p. 102634, Jul. 2024, doi: 10.1016/j.ecoinf.2024.102634.

[35] G. Y. Oukawa, P. Krecl, and A. C. Targino, “Fine-scale modeling of the urban heat island: A comparison of multiple linear regression and random forest approaches,” *Sci. Total Environ.*, vol. 815, p. 152836, Apr. 2022, doi: 10.1016/j.scitotenv.2021.152836.

[36] M. Akintola and G. Neziri, “An Open-Source Deep Learning Framework for Scalable Urban Heat Island Detection Using Geospatial Data,” *Int. Arch. Photogramm. Remote Sens. Spat. Inf. Sci.*, vol. XLVIII-4/W13-2025, pp. 11–16, Jul. 2025, doi: 10.5194/isprs-archives-XLVIII-4-W13-2025-11-2025.

[37] L. Salvadori, M. G. Badas, A. Di Bernardino, G. Querzoli, and S. Ferrari, “A Street Graph-Based Morphometric Characterization of Two Large Urban Areas,” *Sustainability*, vol. 13, no. 3, p. 1025, Jan. 2021, doi: 10.3390/su13031025.

[38] Z. Yu, J. Zhang, G. Yang, and J. Schlaberg, “Reverse Thinking: A New Method from the Graph Perspective for Evaluating and Mitigating Regional Surface Heat Islands,” *Remote Sens.*, vol. 13, no. 6, p. 1127, Jan. 2021, doi: 10.3390/rs13061127.

[39] L. Ruiz, F. Gama, and A. Ribeiro, “Graph Neural Networks: Architectures, Stability and Transferability,” Jan. 29, 2021, *arXiv*: arXiv:2008.01767. doi: 10.48550/arXiv.2008.01767.

[40] S. Yao and B. Huang, “Spatiotemporal Interpolation Using Graph Neural Network,” *Ann. Am. Assoc. Geogr.*, vol. 113, no. 8, pp. 1856–1877, Sep. 2023, doi: 10.1080/24694452.2023.2206469.

[41] L. Shao, Z. Zhou, Y. Sun, J. Xie, and C. Hong, “A novel GCN-based framework for uncovering diurnal patterns and drivers of the surface urban heat island (SUHI) in hyper-dense cities,” *Appl. Geogr.*, vol. 191, p. 103993, Jun. 2026, doi: 10.1016/j.apgeog.2026.103993.

[42] H. Li *et al.*, “Spatiotemporal Graph Learning on Urban Environments,” *Environ. Sci. Technol.*, vol. 60, no. 7, pp. 5191–5207, Feb. 2026, doi: 10.1021/acs.est.5c12640.

[43] S. Takedomi *et al.*, “High-Resolution Forecasting of Road Surface Temperature Using Graph Neural Networks,” in *2025 IEEE 28th International Conference on Intelligent Transportation Systems (ITSC)*, Nov. 2025, pp. 2173–2180. doi: 10.1109/ITSC60802.2025.11423457.

[44] G. Boeing, “OSMnx: New Methods for Acquiring, Constructing, Analyzing, and Visualizing Complex Street Networks,” *Comput. Environ. Urban Syst.*, vol. 65, pp. 126–139, Sep. 2017, doi: 10.1016/j.compenvurbsys.2017.05.004.

[45] Y. Chen, W. S. Afandi, D. Gura, and Y. Kosenok, “A unified deep learning framework integrating OpenStreetMap for multi-domain urban planning tasks,” *Sci. Rep.*, vol. 16, no. 1, p. 10870, Mar. 2026, doi: 10.1038/s41598-026-45927-x.

[46] S. Hochreiter and J. Schmidhuber, “Long Short-Term Memory,” *Neural Comput.*, vol. 9, pp. 1735–1780, Nov. 1997, doi: 10.1162/neco.1997.9.8.1735.

[47] W. L. Hamilton, R. Ying, and J. Leskovec, “Inductive Representation Learning on Large Graphs,” Sep. 10, 2018, *arXiv*: arXiv:1706.02216. doi: 10.48550/arXiv.1706.02216.